When Should a Sim-Pretrained World-Model Planner Trust Its Imagination?

EE535/AI535 Final Project Proposal

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I. Problem

SimDist (RSS 2026, arXiv:2603.15759) pretrains a latent world model in simulation and controls a quadruped with a sampling planner (MPPI) that imagines hundreds of action sequences and executes the best-scored one. On the real robot the model is never exactly right, and a planner is the worst consumer of a wrong model: it searches for the highest score, so it picks candidates whose score is *inflated* by model error. Existing remedies widen coverage or penalise uncertainty, which makes the robot stop rather than fall less. This project asks: *can the planner learn, from its own track record, when its imagination is trustworthy, and plan only then?*

II. Preliminary measurements

Sim-to-sim benchmark (Isaac Lab, Go1, 6 seeds $\times$ 3 speeds); metric: falls per 1000 control steps (lower is better). (1) **Planning helps inside the model's coverage and hurts outside it.** With a model fine-tuned on broad domain randomisation (motor 0.3–1.1, friction 0.2–1.2), prior-only vs. planning gives $0.46 \rightarrow 0.20$ falls at motor 0.4 (inside coverage) but $0.03 \rightarrow 0.47$ at friction 0.15 (outside; CI $[0.09, 0.81]$, progress $55 \rightarrow 33$ m), all extra falls at 0.9 m/s. Writing the planner's gain as *promised gain* minus *selection bias* predicts this (the return-error spread is 0.02 at 0.3 m/s, 0.16 at 0.9 m/s). (2) **Selection bias grows with search.** Raising the MPPI budget from 450×8 to 1000×16 samples$\times$iterations raises falls from $0.47 \rightarrow 5.02$ at friction 0.15 and $0.20 \rightarrow 1.66$ at motor 0.4: a stronger optimiser exploits the same model error harder. (3) **Adaptation moves the boundary, it does not remove it.** With the narrow (paper) model, adapting the dynamics on low-friction data makes planning helpful on friction ($1.16 \rightarrow 0.22$) but not on motor 0.4 ($3.23 \rightarrow 4.14$); adapting on weak-motor data gives the mirror image (motor $3.23 \rightarrow 2.33$, friction $1.16 \rightarrow 5.41$). (4) **Errors predict risk, but the history already knows.** The model's one-step errors are persistent (autocorrelation 0.88/0.58/0.42 at lags 1/25/100) and a 25-step error window predicts a fall within 100 steps with AUC 0.71–0.98; but a probe on the state–action history the model already sees scores 0.80–0.98 and identifies the physics with 0.99 accuracy. The information is in the history; what is missing is a score *calibrated* to what this model actually achieves.

III. Method

The SimDist core (encoder, history encoder, dynamics, reward, value, prior π) stays *frozen*; two parts are added. **(a) Calibrated outcome head.** Given the history tokens the model already computes, a candidate action chunk and its promised return $\hat{J}$, a head outputs quantiles of the *realised* 25-step return and the probability of a fall within 100 steps. Unlike the frozen value head (labelled by the expert's value) it is labelled by what this model's own planner obtained, so it absorbs the model's error on the current physics and the planner's selection bias; MPPI scores candidates with a lower quantile instead of $\hat{J}$. **(b) Plan-on-demand gate.** Each step the prior's own action chunk is scored by (a); if it is safe the prior is executed without planning, otherwise MPPI runs with calibrated scores, blended with the prior in proportion to the predicted improvement. **Training.** The frozen model is run in deliberately mismatched simulated physics (motor, friction, payload per episode; half prior-only, half planning), recording each plan's imagined return and what actually happened; realised returns and falls are free labels, and (a) is trained by quantile regression with one physics axis held out. No weights change at deployment. A variant of (a) that also receives the last 25 prediction errors is kept only as an ablation (measurement 4).

IV. Experiments and metrics

Simulation (main). Conditions: nominal, motor 0.4/0.5, friction 0.15, payload 3 kg and a 0.4$\rightarrow$0.8 motor switch, for a narrow and a broad pretrained model, zero-shot and after adaptation. Arms: prior-only, MPPI, MPPI+(a), MPPI+(a) with error tokens (ablation), gate (b), an ensemble-uncertainty penalty and our earlier norm-threshold gate (baselines). Metrics: falls per 1000 steps (paired bootstrap CI over 18 runs), reward per step, progress, tracking error, planning fraction ρ (latency $\approx 48\rho + 3(1-\rho)$ ms). Pre-registered gates: on a held-out physics axis the calibrated head ranks candidates by realised return better than $\hat{J}$ (Spearman ≥ 0.5 vs. 0.36 measured for $\hat{J}$); the gate is within CI of the better of prior-only and MPPI in every condition and better than both in at least one, also after adaptation (measurement 3). **Hardware (Nov. 10–24).** Real Go1: nominal, PTFE sheet, 3 kg payload; prior-only vs. planner vs. gate, 10 trials each, before and after 30 minutes of adaptation; success over 10 m, progress, tracking error, ρ; predicted pattern = measurements (1) and (3).

Part of work intended for RSS 2027; SimDist is by other authors and the final report will include the original paper, as required.